



OPEN Exploring neural markers of dereification in meditation based on EEG and personalized models of electrophysiological brain states

Tamas Madl

With mounting evidence for the benefits of meditation, there has been a growing interest in measuring and quantifying meditative states. This study introduces the Inner Dereification Index (IDI), a class of personalized models designed to quantify the distance from non-meditative states such as mind wandering based on a single individual's neural activity. In addition to demonstrating high classification accuracy (median AUC: 0.996) at distinguishing meditation from thinking states moment by moment, IDI can accurately stratify meditator cohorts by experience, and correctly identify the practices most effective at training the dereification aspect of meditation (decentering from immersion with thoughts and perceptions and recognizing them as mental constructs). These results suggest that IDI models may be a useful real-time proxy for dereification and meditation progress, requiring only 1 min of mind wandering data (and no meditation data) during model training. Thus, they show promise for applications such as real-time meditation feedback, progress tracking, personalization of practices, and potential therapeutic applications of neurofeedback-assisted generation of positive states of consciousness.

Keywords Meditation, EEG, Mindfulness, Neurofeedback, Dereification, Modes of existential awareness

Exploration of human consciousness is one of the frontiers of basic neuroscience, and many experimental and quantitative tools have been applied to the attempt. Structural changes associated with mindfulness meditation were identified in brain areas involved in attention control (anterior cingulate cortex—ACC—and striatum), emotion regulation (prefrontal regions, limbic regions, striatum), and self-awareness (insula, medial prefrontal cortex, posterior cingulate cortex, and precuneus)^{1,2}. Such results were mainly based on brain imaging methods such as fMRI, which require expensive and intrusive equipment and are limited in temporal resolution.

To investigate electrical activity on shorter time scales, electroencephalographic (EEG) studies were also carried out, and report a range of changes associated with various different meditative states. Notably, they observe increased power in the slow-wave bands such as theta (3–7 Hz) and alpha (8–12 Hz) frequencies of cortical activity³. Additionally, there are reports of increased entropy and complexity^{4,5}, along with heightened cortical coherence and synchrony across different brain regions⁶. Some studies have reported contrasting findings, such as decreases in alpha power⁷ or reductions in complexity^{8–10} and synchrony^{11–13}, likely due to the diversity and heterogeneity of the investigated meditation styles and associated brain activity patterns.

These markers are limited in their specificity and predictive accuracy with regard to distinguishing between states of consciousness in individual practitioners. First, there is a high amount of inter-subject variability of EEG measures^{14–16}. In fact, cortical activation patterns are individual enough that features extracted from delta, theta, alpha, beta and gamma bands can be used for accurate biometric identification of individuals^{17,18}, implying that personalized models may characterise neural activity more accurately than the attempt to generalize electrophysiological markers across individuals.

Second, much of the research on markers of meditative or altered states of consciousness focuses on descriptive statistics, rather than attempting accurate intra-subject prediction in real time. For example, a recent systematic review¹⁹ cites dozens of computational studies, but none of them report intra-subject classification performance. Some more recent studies do perform classification of meditative practice states, attempting to generalize across subjects and predict states within subjects. However, they tend to report limited accuracies (e.g. 74% with SVM²⁰, 73–78% with multilayer perceptrons²¹, 78–84% with decision trees, boosting, and random forests²²) or to require long sliding windows (e.g. 30 s²³–60 s²⁴) as well as long recordings from both meditative practice states

Austrian Research Institute for Artificial Intelligence (OFAI), Wien, Austria. email: tamas.madl@gmail.com

and non-meditative states, making them impractical for real-time applications such as neurofeedback (Note: In this paper, we use the phrase ‘meditative practice states’ to refer to states participants enter into when encouraged to practice the meditation method of their tradition. The phrase is not meant to imply the existence of a single, uniform, homogeneous state. Meditative states vary greatly across participants and meditation styles, reflecting the diversity and heterogeneity of practices across different traditions)

To address these limitations, we propose a novel class of methods for personalized learning from minimal subject-specific data, enabling state quantification without requiring subjects to master a certain desirable state. This ability to learn from a single data category allows the method presented below to be used on data from novice meditators, for practices without a fixed ‘end goal’, for neurofeedback, and in many other settings where substantial data on a target state may be unavailable—unlike existing work on trying to automatically acquire models of meditative states using machine learning^{19–27}, which all require data from at least two different states (e.g. an undesirable and a desirable state) to be trained and make predictions.

Our approach, which we term the Inner Dereification Index (IDI), employs one-class classification/outlier detection techniques from machine learning literature^{28–30}. We propose that this method can serve as an approximate, partial, and limited proxy for dereification, particularly when applied to brain regions corresponding to the DMN. It is important to note that the IDI is intended as an exploratory tool for approximating shifts in awareness associated with meditation, not as a measure of a single well-defined process (see limitations in “Discussion” section).

Deviations from default mode network activity as computational markers of meditative practice states

The primary rationale for focusing on deviations from the mind-wandering state as indicators of meditation is based on robust evidence linking meditation practices with the downregulation of the default mode network (DMN), a network of brain regions strongly associated with self-referential processing, mind-wandering, and cognitive fusion. Numerous studies have shown that meditation, particularly practices aimed at dereification or non-attachment to thoughts, significantly reduce DMN activity^{31–35}. Furthermore, the degree of DMN downregulation seems to correlate with meditation experience^{36,37}. This relationship is also observed in other altered states of consciousness, such as those induced by psychedelics, where reduced DMN activity is associated with profound changes in the substantiality, solidity and contraction of the sense of self³⁸. Given these findings, it is plausible to hypothesize that departures from default mode activity may correlate with key aspects of the meditative state such as dereification. Our methodology builds on this hypothesis by identifying EEG patterns that deviate from those associated with mind-wandering, using them as potential indicators of meditation-induced shifts in consciousness.

Below, we focus on dereification as a potential experiential correlate of reduced DMN activity, and thus of the quantitative method presented in this paper. Dereification refers to the experiential awareness of the distinction between actual objects and their conceptual representations, and encompasses a gradient, most clearly articulated by the framework proposed by Dorjee³⁹ (who builds on earlier work by⁴⁰—even earlier works have referred to the same concept as ‘phenomenological reduction’⁴¹ or simply ‘mindful attention’⁴², although the latter terms have been used with less precision).

- The lower end, minimal dereification, is characterised by full immersion in mental phenomena (fusion with the contents of consciousness), leading to a diminished capacity to monitor and/or regulate psychological processes.
- Decentering, more advanced on the gradient, involves recognizing thoughts and feelings as as fleeting mental phenomena rather than unchangeable facts. It’s characterized by a mindful awareness where internal experiences are seen and accepted without over-identification, facilitating emotional balance and reduced reactivity. This perspective shift, fundamental in mindfulness-based therapies such as MBCT⁴³, has been shown to be therapeutically significant^{44–46}.
- The higher end of the dereification gradient involves fully recognizing and experiencing thoughts, feelings, perceptions, and the sense of self as mere mental constructs devoid of intrinsic, separate existence, rather than concrete, unchangeable realities—often referred to as ‘emptiness’ in Buddhist-inspired practices⁴⁷. As one of the limitations of this view (more in “Discussion” section) it is important to point out that there is no simple one dimensional relationship between mind-wandering, DMN activity, and dereification. Not all DMN activity equates to mind-wandering^{48,49}, not all mind-wandering is mediated solely by the DMN⁵⁰, and advanced meditators may experience high levels of dereification even with some DMN activation⁵¹ although the two tend to be anti-correlated³³. Finally, there are states that are consistent with a down-regulated DMN but would not be considered a dereified state by most traditions, such as dreamless sleep.

Nevertheless, we propose the approximation of DMN downregulation as roughly corresponding to dereification as a tentative quantitative correlate of meditative practice states, for three main reasons. First, decentering and dereification have repeatedly been shown to be a significant and central aspect of mindfulness^{52–54}, and implicated in a wide range of contemplative practices including attentional, constructive as well as deconstructive families of meditation^{55–57}. Second, as mentioned above, DMN areas are some of the most significant neural correlates of meditative states^{31–35} and meditation experience^{36,37}. Finally, although there are other components of meditation described in literature, dereification and decentering have been emphasized by several scholars^{39,55,58} as fundamental to the shifts in the awareness of self and reality enabled by meditation (some authors make stronger claims, e.g.: ‘all styles of mindfulness meditation train this capacity deliberately’⁵⁸ or ‘[the first steps of] dereification [are] dereification are foundational for more “advanced” practices’⁵⁹). They have also been claimed to play a crucial role in psychological health^{46,60,61}.

We propose that our model score can be used as a proxy variable for dereification, since it measures the distance from electrophysiological brain states characterised by minimal dereification, namely immersion into conceptual thought or mind wandering. Consequently, this measure could approximate the progression away from the lower end of the dereification gradient.

In the following sections, we empirically investigate the properties of the IDI in quantifying brain states characterized by both the lowest and the highest ends of the dereification gradient. We apply the IDI to complete meditation beginners, intermediate practitioners, and experienced meditators specifically aiming for the highest levels of dereification in their practice.

Furthermore, we validate the IDI with advanced Tibetan practices (Datasets 3 and 4) and mindfulness on psilocybin (Dataset 5) to investigate its robustness as a measure of dereification across both traditional meditation and altered states of consciousness.

Methods

Subjects and data

For the meditation experiments (“[Estimated dereification is strongly associated with meditation experience](#)” and “[Prediction accuracy: meditation vs. mind wandering](#)” sections) we have used five sets of online open-access electrophysiology data, from a total of 220 participants, ranging from zero hours to over 30,000 h of meditation experience. For dataset 1 (beginning and intermediate meditators), we considered the dataset made available by⁶², 58 participants recruited from meditation centers in Belgium and predominantly practicing mindfulness meditation (except 8 Zen and 9 Vipassana practitioners). All participants were instructed to do breath mindfulness in the meditation condition. Two groups were recruited, 29 participants (17 females) with at least three years of meditation experience (intermediate meditators) with ages of 47.31 years on average (SD 11.21), and the other 29 (15 females) with no reported previous meditation experience (beginners) aged 47.13 (SD 13.93) years on average. Data from this group was recorded using a Nexus-32 EEG system, a 19 electrode cap (plus two reference electrodes and a ground electrode) positioned in accordance to the 10-20 system.

For dataset 2, data from the Meditation Research Institute in Rishikesh was used, collected and made openly available by⁶³. This group was comprised of 20 meditators from an open awareness/“doing nothing” meditation tradition (Isha Shoonya Yoga—SNY, aged 40 ± 10 years, two females, 2 625+/-1 868 hours experience), a tradition combining open monitoring and focused attention on somatosensory sensations (Vipassana—VIP, aged 47+/-15 years, two females, 9201+/-7759), and 27 meditators from a focused meditation tradition (Himalayan Yoga Tradition, HYT, aged 43+/-12 years, two females, 15 475+/-12 748 hours experience). Additionally, 32 healthy control individuals without prior meditation training (45+/-10 years, 5 females, 0 h experience) were recruited. EEG was recorded by⁶³ using a 64 channel Biosemi Active-Two system with 10-20 electrode placement.

We have also included two datasets in order to specifically investigate the claim that our computational metric may correlate with dereified modes of existential awareness. Dataset 3 was collected by⁵, sourced from Dzogchen Trekchö practitioners. Trekchö means ‘cutting through’ and is a practice within Tibetan Dzogchen Buddhism that focuses on piercing through the illusions of reified thoughts, perceptions, and self to reveal the mind’s innate emptiness and clarity^{64–66}. Dataset 3 contained 5 advanced Dzogchen meditators (1 female) with 36 028+/-17 478 hours of lifetime practice⁵.

Dataset 4 comprised data from Tibetan Guhyasamaja practitioners (recorded and shared by⁶⁷), to provide further empirical evidence of our model’s ability to constitute a proxy for dereification. Guhyasamaja, a foundational tantric practice in Tibetan Buddhism, involves a progressive meditation culminating in the ‘clear light’ stage—a state characterized by abiding at the subtlest level of mental activity, naturally free from conceptual cognition and grasping for true existence. This cohort consisted of 30 monks with at least a decade of practice (est. 20,000 h), each successfully passing through the eight stages of Guhyasamaja meditation during EEG recording with the NVX-52 acquisition system (Medical Computer Systems Ltd., Moscow, Russia) at a 500-Hz sampling rate. This addition complements Dataset 3 (Dzogchen Trekchö practitioners), both datasets offering valuable insights into meditation techniques directly aimed at dereification and insights into the emptiness of the nature of mind, albeit through different doctrinal approaches.

Finally, we included Dataset 5, a cohort of individuals with little or no meditation training performing breath mindfulness under the influence of psilocybin. The data was collected and made available by⁶⁸ and includes 58 participants, all over the age of 18, recruited online through the Psynautics project. Due to the strict anonymity required by a study involving controlled substances, no further details on the participants is available. EEG was recorded by participants at home with a consumer device, Interaxon Muse^{69,70}, through the app MindMonitor. Data from 10 participants was excluded due to the recording being either too short (< 1 min) or having too low sampling rate (< 200 Hz sampling rate).

Procedure

All datasets contain recordings of mind wandering states as well as meditation states from all participants.

In dataset 1⁶², participants were encouraged to let their mind wander for a block of 5 min, and encouraged to meditate for a block of 10 min, with breath-focused mindfulness meditation instructions taken from⁷¹. This was the least clearly specified mind wandering states of the three cohorts, but the relative inexperience of meditators in this cohort makes it unlikely that they would have spontaneously fall into meditation when setting the intention to let their mind wander.

In dataset 2⁶³, the experimental procedure consisted of two 20-min sessions, a “meditation” block (consisting of 10 min breath focus meditation and 10 min meditation in the participant’s tradition), and an “instructed mind wandering” block.

In ‘Instructed Mind Wandering’ (IMW), they focused on personal memories: they were explicitly instructed to perform mind-wandering for 20 min per block, by recalling real autobiographical events which were

emotionally neutral such as routine childhood life, travels, etc. from a first-person perspective. This kind of mind wandering—focused on personal memories and experiences—typically involves a first-person self representation and immersion in conceptual, autobiographical thought, and tends to involve minimal or no dereification (at a minimum, it relies on language—linguistic/conceptual thought—which involves some degree of reification of and duality between subjects and objects being contemplated).

Datasets 3 used a similar Instructed Mind Wandering condition as above, but shorter (80 seconds per participant in dataset 3), and focused on recent personal memories (*'think about your day since the moment you woke up this morning'*) instead of encouraging earlier memories such as childhood life or travels. In either case, mind wandering condition was characterized by self-directed thoughts of the past (consistent with literature on the topic^{72,73}), established a rest condition for training the models that is less heterogeneous than unstructured resting, and aimed to prevent highly skilled meditators from automatically engaging in their meditative practice during the non-meditative condition (and thus minimize the application of dereification skills to their experience).

Although Dataset 4 did not include explicit mind wandering instructions, the first step in the Guhyasama meditation procedure involves effortful visualization practices (form, colors, symbols, and attributes of deities) which can constitute a control condition with similar properties. The meditation condition commenced directly after the mind wandering / visualization block, and lasted 600 seconds in dataset 3, and determined by the meditators (on average, 36min+/-SD 16min) in dataset 4.

Finally, in Dataset 5 (psilocybin), the control condition was a recording of everyday consciousness (resting state) prior to taking any mind-altering substances, while the meditation condition involved breath mindfulness carried out while under the influence of psilocybin, for a duration chosen by participants (317+/-SD 113 seconds on average).

EEG preprocessing

For each cohort, EEG data underwent a series of standard pre-processing steps. Initially, we applied a notch filter, set at 50 Hz or 60 Hz depending on the region's power line frequency (50 Hz for Europe and India in datasets 1 and 2, and 60 Hz for the two US participants in dataset 3), to remove electrical noise interference. Subsequently, we conducted Independent Component Analysis (ICA)⁷⁴, to identify and remove artifact components related to eye blinks, muscle artifacts, heart beat, or line or channel noise⁷⁵.

As the final pre-processing step, we computed Time-Frequency Representations (TFR) using Morlet wavelets⁷⁶ on selected channels of the EEG data. This approach was implemented using the MNE Python package for neuroscience data analysis⁷⁷. Unlike the Short-Time Window Fourier Transform which has a fixed window size throughout, wavelets allow a signal to be analyzed at different frequencies with different resolutions. Thus, wavelets can facilitate better temporal resolution at high frequencies, which is important especially for short window sizes and high frequency ranges considered in this paper.

Our focus for the Morlet-based TFR was on the high gamma band, specifically spanning 16 bands of 1 Hz each between 45 and 60 Hz. This frequency range was selected based on recent findings underscoring the significance of high gamma power in studies exploring advanced states of dereification⁷⁸. Additionally, this band range aligns with the findings from authors of dataset 2, who reported notable changes in high gamma activity during meditation⁶³. The choice to concentrate on this particular frequency range is also supported by literature reviews highlighting the relevance of high gamma oscillations in cognitive and contemplative processes^{79,80}.

For the meditation experiments, we consider only a subset of electrodes which have been argued to be most strongly associated with desirable meditative states, including diminished self-reference and reduced mind wandering⁷⁸. These include the 10-20 electrodes in closest spatial proximity to areas of the Default Mode Network (DMN), which are often linked to Brodmann areas (BA) 7, 10, 23, 25, 31, and 32, as well as the Anterior Cingulate Cortex (ACC; BA 24, 32, and 33), the Posterior Cingulate Cortex (PCC; BA 23, 31), the Precuneus, and the superior parietal lobule (BA 7 and 5). We used the Van Essen Lab's Population-average, Landmark- and Surface-Based (PALS-B12) of the human cerebral cortex⁸¹, mapped to FreeSurfer's^{82,83} fsaverage subject, to identify the most suitable electrode locations. The electrodes with the closest spatial proximity P1 (15 mm) and P2 (15 mm) for BA 7 L and R, Fp1 (14 mm) and Fp2 (16 mm) for BA 10 L and R, Pz for BA 23 (43 mm/45 mm L and R), F1 (36 mm) and F2 (39 mm) for BA24 L and R, Pz for BA 31 (36 mm/33 mm L and R), and AFz and F1 for BA 32 (33 mm/33 mm L and R). These electrodes significantly overlap with previously proposed placements and regions of interest in studies exploring DMN⁸⁴ and meditation^{78,85,86}.

We recognize the imprecision in using scalp electrodes proximal to the cingulate cortices, which are situated inter-hemispherically and over 3cm beneath the scalp surface. Despite this shortcoming, in our experiments, the above-mentioned electrodes led to the highest discriminative power between meditative practice states. Alternative approaches such as the application of source localization techniques led to a decrease in accuracy compared to the use of data from electrodes positioned closest to these areas (as did omitting data from electrodes closest to cingulate cortices).

Table 1 provides a summary of the preprocessing pipeline's output. For model training, we selected a random 1-min segment from the mind-wandering EEG data, which was then divided into 3-second windows. These segments were transformed into 128-valued vectors using Morlet Time-Frequency Representation (TFR) on 8 channels and 16 frequency bands, as described above, and served as the input for model training. During the evaluation phase, we utilized the remaining data for performance analysis—this included the mind-wandering data (excluding the 1-min training subset) and all meditation data.

Computational methods

To achieve the objectives outlined in the introduction—creating subject-specific models, requiring minimal training data (1 min or less), and applicability without needing data on target meditative practice states—we

Brodman area	Hemi-sphere	Closest Chan-nel	16 gamma frequency bands, 1 Hz wide, 45–60 Hz
BA7	L	P1	A flattened 8(channels)x16(frequency bands)=128 valued vector from TFR calculated on pre-processed 3s EEG time window serves as input for IDI algorithm
	R	P2	
BA10	L	Fp1	
	R	Fp2	
BA23	L,R	Pz	
BA24	L	F1	
	R	F2	
BA31	L,R	Pz ^a	
BA32	L	F1 ^a	
	R	AFz	

Table 1. Summary of channels and frequency bands used as input for the IDI algorithm. ^aDuplicate channels (after deduplication, 8 channels in closest spatial proximity to areas of the Default Mode Network are used).

employed one-class classification/outlier detection techniques from machine learning literature^{28–30}. (Note: the phrase ‘meditative practice states’ is a shorthand and may refer to a subset of a vast set of diverse, dereified states of consciousness targeted by various meditation systems, and is by no means indicative of a singular, uniform, homogeneous state or goal. For a more detailed description of dereification—a key feature of meditative practice states in many traditions—see “[Deviations from default mode network activity as computational markers of meditative practice states](#)” section). This approach is fundamentally different from existing statistical approaches for finding biomarkers of meditation (see Introduction), as well as methods based on modern machine learning to learn binary classifiers distinguishing meditative practice states from non-meditative states (e.g.^{19,25–27,87}). Contrastingly, traditional methods leverage datasets from many participants, encompassing all states (both meditative and mind wandering in our case). They employ statistical learning to create inter-subject classifiers aimed at generalizing across participants, rather than tailoring to individual patterns. One of the most popular class of machine learning methods for binary classification is based on gradient-boosted decision trees (combining many weak decision tree models to an ensemble, and iteratively learning new models to compensate for errors, by fitting new decision trees to the residual errors patterns to reduce overall error). This class of methods includes XGBoost⁸⁸ and LightGBM⁸⁹, where the latter has been shown to achieve significantly higher performance on most datasets and applications^{89,90}.

Figure 1 illustrates the difference between the binary classification model requiring many subjects and data from meditative practice states (left) to the class of anomaly detection/one class classification methods which are the focus of the current work, which allow learning from a single class, without data on the meditative states (right).

A wide range of anomaly detection/one class classification methods have been proposed in literature, and recently evaluated systematically on a diverse set of data by⁹¹, who group the proposed methods into the following categories:

- probabilistic (sampling or distribution based methods),
- linear (methods based on linear projection or approximation),
- proximity based (methods based on the spatial distance or cluster structure of data points), as well as
- ensemble (combinations of multiple methods), and
- graph and neural network methods. Here, we have chosen the best-performing method from each category except the latter two, and ensembled the models (averaging their normalized predictions) to provide the final estimate. We have excluded the graph and neural network categories due to their higher computational complexity, larger parameter space, and need for hyperparameter tuning (stable hyperparameter tuning is almost impossible with the very small dataset sizes implied by the requirement to train the models on 1 min of EEG or less—max. 20 data points in the meditation experiment results of “[Estimated dereification is strongly associated with meditation experience](#)” and “[Prediction accuracy: meditation vs. mind wandering](#)” section, which use 3 second windows). The methods we have chosen tend to excel on very small training datasets (see⁹¹ for an extensive benchmark).

We only include the best-performing method from each remaining category. Our reasons include that error patters of various models in the same categories tend to be correlated, that ensembles tend to only benefit from models with uncorrelated error patterns, as observed in literature^{92–94} as well as in our data (trying to include additional models of the same category—rather than only the best-performing model of each category—in the ensemble does not improve, and often decreases, predictive accuracy on our data).

Specifically, we have used an averaging ensemble of the following methods, which calculate an ‘outlierness score’ $q(\mathbf{x}, \mathcal{X})$ (ideally yielding small values for inliers, and large values for outliers, based on a particular data instance $\mathbf{x} \in \mathbb{R}^D$ and a dataset $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N | \mathbf{x}_i \in \mathbb{R}^D\}$ consisting of N data points of dimensionality D):

- Linear: the PCA-based algorithm described in⁹⁵, which defines the outlierness q of a data point as its distance in the principal component space. That is,

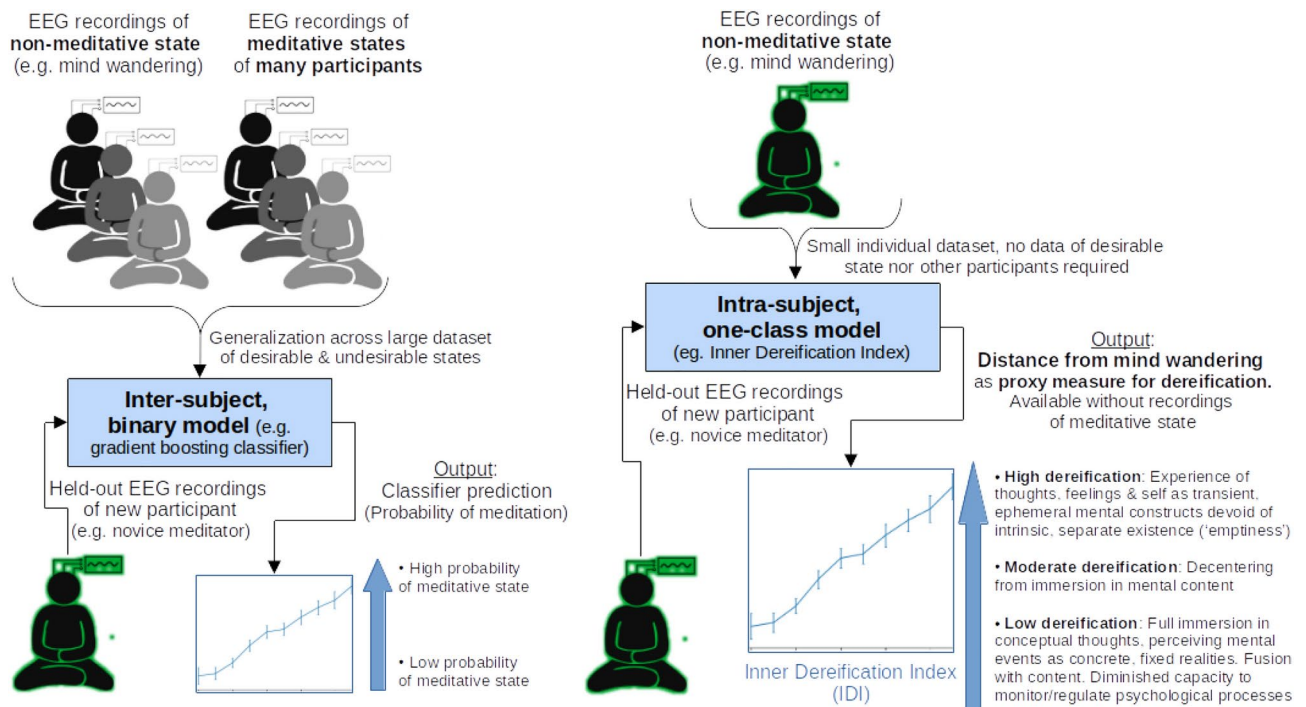


Fig. 1. Two approaches for classifying and quantifying meditative states. Left (binary classification): inter-subject learning methods, e.g. machine learning classifiers such as gradient boosted decision trees, can be used in settings where a large amount of data is available from meditative practice states and mind wandering states. This approach has the disadvantage of needing data from both kinds of states, requiring patterns that can be generalised across participants (whose cortical patterns and neural architecture may differ, reducing the accuracy of generalised inference), and producing probabilities rather than proxies or correlates of aspects of the meditative state. Right (one-class models): one-class learning methods or outlier detection approaches can be used in settings where data is available only from non-meditative states (with low dereification), such as mind wandering. This approach has the advantage of not needing a large cohort of other participants, being able to learn from the data of a single participant, even if there is no data available characterising target meditative states (e.g. novice meditators, or meditators aiming for states they have not experienced before). It also provides a quantitative metric that may be used to compare or measure progress across or within meditation sessions (e.g. neurofeedback), and may constitute an approximate, partial and limited proxy for dereification.

$$q_{PCA}(\mathbf{x}) = \sum_{i=1}^D \frac{y_i^2}{\lambda_i} \quad (1)$$

for D -dimensional data point $\mathbf{x} \in \mathbb{R}^D$, where y_i denotes principal component i of sample $\mathbf{x}$, and λ_i the sample variance of principal component i .

- Probabilistic: the simple *sampling*-based algorithm of⁹⁶, which randomly and independently samples a subset $S(\mathcal{X}) \subset \mathcal{X}$ only once, and the outlieriness score q is defined (based on the distance d from the closest data point in the subset sample) as

$$q_{\text{sampling}}(\mathbf{x}, \mathcal{X}) := \min_{\mathbf{x}' \in S(\mathcal{X})} d(\mathbf{x}, \mathbf{x}') \quad (2)$$

Despite the simplicity of this approach, it has been claimed by the original authors to significantly outperform many state-of-the-art techniques, including One-Class Support Vector Machines (SVMs)⁹⁷, Isolation Forests⁹⁸, and many outlier factor scores—see⁹⁶ -, a claim that also held on our data.

- Proximity: the *cluster*-based algorithm of⁹⁹, which first performs K-Means clustering on the input dataset, and, intuitively, tries to define outliers from the point of view of clusters, as data points that do not lie in any large clusters.

$$q_{cluster}(\mathbf{x}) = \begin{cases} |C_i^*| \min_{j \in \{1 \dots b\}} d(\mathbf{x}, C_j), & \text{where } \mathbf{x} \in C_i, C_i \in SC, C_j \in LC \\ |C_i^*| d(\mathbf{x}, C_i) & \text{where } \mathbf{x} \in C_i \text{ and } C_i \in LC, \end{cases} \quad (3)$$

Where $|C_i|$ is the size of cluster i . Thus, the outlierness of a record is determined by the size of its cluster, and the distance between the record and its closest cluster (if it lies in a small cluster) or the distance between the record and the cluster it belongs to (if the record belongs to a large cluster). The algorithm is parameterized by the boundary between large clusters LC and small clusters SC, as well as the number of clusters k .

The final outlierness score is calculated by ensembling the normalized scores. Normalization is necessary to render the scores onto roughly the same scale and thus facilitate ensembling by averaging (since each of the methods produce outlier scores on different scales, and may take values up to e.g. the minimum inter-cluster distance for $q_{cluster}$, or the sum of squares of principal components divided by sample variances for q_{PCA}). To avoid overfitting and bad generalization to unseen data, normalization scores are estimated using a K-fold cross-validation procedure. The input data $\mathcal{X}$ is partitioned into K folds: $\mathcal{X}_1, \dots, \mathcal{X}_K$, and for each method above, the maximal outlierness on the held-out data is calculated, e.g.:

$$M_{PCA}(\mathbf{x}, \mathcal{X}) = \max_{i \in \{1 \dots |\mathcal{X} \setminus \mathcal{X}_k|\}, k \in \{1 \dots K\}} q_{PCA}(\mathbf{x}_i \in \mathcal{X} \setminus \mathcal{X}_k, \mathcal{X}_k) \quad (4)$$

Normalizing method-specific outlierness scores with their respective maximum based on a held-out set (as defined by Eq. (4)) and averaging the normalized scores yields the final ensemble score, defined in Eq. (5). (Note: although normalized using the training dataset, IDI scores may exceed 1 on the test set. If the latter contains significant outliers—often the case when applying IDI trained on mind wandering and tested on meditative practice states—, IDI scores and subscores tend to be significantly larger than 1 on the test set.)

$$q_{ensemble}(\mathbf{x}, \mathcal{X}) = \frac{1}{3} \left(\frac{q_{PCA}(\mathbf{x}, \mathcal{X})}{M_{PCA}} + \frac{q_{sample}(\mathbf{x}, \mathcal{X})}{M_{sample}} + \frac{q_{cluster}(\mathbf{x}, \mathcal{X})}{M_{cluster}} \right) \quad (5)$$

Finally, since the differences between meditative and non-meditative states can become very large in the case of experienced practitioners with tens of thousands of hours of practice, we take the logarithm of the ensemble score to arrive at the *Inner Dereification Index* (IDI):

$$IDI(\mathbf{x}, \mathcal{X}) = \log_2(2 + q_{ensemble}(\mathbf{x}, \mathcal{X})) \quad (6)$$

To obtain this score, the three one-class models underlying $q_{ensemble}$ have to be trained and ensembled. The ensemble makes prediction on individual data points $\mathbf{x}_i \in \mathcal{X}$ possible, e.g. on short temporal slices of an EEG recording during meditation.

Preprocessing. EEG data is detrended, notch filtered, and stripped of artifacts with ICA. A random 1-min subset of the mind wandering data is extracted, and sliced into 3 second windows. Average power is computed (Time-Frequency Representation using Morlet wavelets) in 16 of 1 Hz gamma frequency bands between 45 and 60 Hz across 8 EEG electrodes most relevant to the Default Mode Network (the major neural correlate of mind wandering), and 20 resulting vectors of $16 \times 8 = 128$ average power values obtained as the training dataset for the one class classifiers: $\mathcal{X}_{train} = (x_1, \dots, x_{20})$, such that $x_i \in \mathcal{R}^{128}$. See “[EEG preprocessing](#)” section for details of the EEG preprocessing pipeline.

Training In each of the experiments below, models were trained only on data from 1-min subsets of mind wandering states, $\mathcal{X}_{train}$, preprocessed as above (no data from any meditative practice state is required or used). All three component models— $q_{PCA}(\mathbf{x}, \mathcal{X}_{train})$, $q_{sample}(\mathbf{x}, \mathcal{X}_{train})$, $q_{cluster}(\mathbf{x}, \mathcal{X}_{train})$ —were first trained on the entire training set (which involves calculating principal components for q_{PCA} , optionally setting up a KDTree for faster distance calculations for $q_{sampling}$, and performing K-Means clustering for $q_{cluster}$ —see¹⁰⁰ for implementation details). Training hyperparameters were set heuristically—all principal components were used for q_{PCA} , 95% of the data used for q_{sample} (such that $|\mathcal{S}(\mathcal{X})|/|\mathcal{X}| = 0.95$), and the parameters of $q_{cluster}$ were left at their default values as per the implementation of¹⁰⁰. Subsequently, normalization scores were calculated using a K-Fold cross validation procedure as defined by Eq. (4) in order to render the scores roughly onto the same scale¹².

Testing/validation Test data utilized all the data available from a participant except for the 1-min subset used for training (with no overlap—none of the training data was used for testing). Test data was preprocessed as above to obtain $\mathcal{X}_{test} = (x_1, \dots, x_N)$, such that $x_i \in \mathcal{R}^{128}$, and $N = \frac{T}{3}$ if T is the number of seconds of the total EEG recording duration (excluding artifacts removed in the preprocessing stage). Prediction scores were obtained one by one from the TFR representation of each 3 second window in the signal, using Eq. (6): $(IDI(\mathbf{x}_1, \mathcal{X}_{train}), \dots, IDI(\mathbf{x}_N, \mathcal{X}_{train}))$

In order to attempt to quantify the entire meditation session of a practitioner (all moments of meditation, not just one), we also introduce the maximal dereification index

$$IDI_{max}(\mathcal{X}_{test}, \mathcal{X}_{train}) = \max_{\mathbf{x}_i \in \mathcal{X}_{test}} IDI(\mathbf{x}_i, \mathcal{X}_{train}), \quad (7)$$

In the sections below, we compare IDI_{max} across various cohorts of meditators, in order to investigate whether this metric behaves as can be expected of a proxy of dereification (1. IDI_{max} should correlate with lifetime practice hours, and 2. IDI_{max} should be significantly larger than expected based on lifetime practice for practices directly aiming for dereification, and smaller for practices which involve reification).

Additionally, we assess IDI's efficacy in differentiating between meditative and thinking states on a moment-to-moment basis. For this latter evaluation, we utilize the area under the receiver operating characteristic (ROC AUC) statistic. ROC AUC evaluates the performance of the IDI score by plotting its true positive rate (TPR)—the rate of correctly identified meditation moments—against its false positive rate (FPR)—the rate of incorrectly predicted meditation moments. It then integrates this curve to yield an overall performance measure^{101,102}. Significantly, the ROC AUC metric does not require a fixed threshold for binary classification. Instead, it integrates the true positive rate as a function of false positive rate at each threshold, over all possible thresholds from the minimum up until the maximum IDI score. When evaluating classification performance with ROC AUC, we took the mind wandering conditions to be the 0 class and the meditation conditions to be the 1 class in our dataset.

The ROC AUC takes a value of 1 in the case of a perfect classifier (all meditation moments correctly identified, zero thinking moments misclassified), and 0.5 in the case of a zero-utility classifier which makes random predictions instead of predicting a correct pattern.

In all experiments reported below, we have used 10-fold cross-validation, and evaluate model performance on held-out test sets not seen by the model at training time. The gradient boosting (LightGBM⁸⁹) model was allowed to learn from 90% of the training data, and evaluated on the held-out 10%, rotating the subsets until all parts of the data have been used for evaluation.

The exact same test splits and test data were used to evaluate the performance of the LightGBM models and the IDI models in order to facilitate direct comparison of the accuracy statistics. The training procedure was different in two aspects: 1) IDI models were only given access to training data of the mind wandering states recorded from a single participant (rather than both meditative and mind wandering states from all participants), and 2) IDI models were only given access to a 1 min window of the EEG recording of the mind wandering state, randomly extracted from the 90% of the EEG data assigned to the training set (rather than being given the entire training slice). Test sets were equivalent between both methods and contained all of the test data (both meditative and mind wandering states) not used at training time.

Results

Estimated dereification is strongly associated with meditation experience

One way of evaluating the quantification method proposed here is to compare IDI_{max} across all cohorts of meditators (as an approximate, partial and limited proxy for dereification with interesting empirical properties, not asserted to be an adequate model—see “[Deviations from default mode network activity as computational markers of meditative practice states](#)” section). Results across cohorts can be seen in Fig. 2. There are statistically significant differences in meditation dereification as estimated by IDI_{max} across all cohorts, as can be expected if a score meaningfully captures a proxy for dereification (which presumably increases with meditation experience and skill), as confirmed by a one-way analysis of variance (ANOVA) with seven levels, which yields $F(8, 211) = 7.38, p < .001$.

IDI_{max} is strongly associated with lifetime practice:

- Pearson correlation of practice hours and IDI_{max} : $r = 0.943, p < 0.001$
 - Spearman correlation of practice hours and IDI_{max} : $r = 0.958, p < 0.001$
 - R^2 coefficient of determination of a linear model between lifetime practice hours and IDI_{max} : $R^2 = 0.900$
- It can also be interesting to look at the dereification gains, or the actual IDI minus what could be linearly estimated based on lifetime practice hours. For a score that only measures meditation expertise, there should be no difference between styles of practice, and dereification gains would be zero or very small and randomly distributed.

The right panel of Fig. 2 suggests that the Inner Dereification Index (IDI) may measure aspects beyond mere meditation proficiency. This is indicated by two notable deviations.

Firstly, in the Vipassana and Himalayan yoga cohorts, which practices focused attention on various body parts, we notice a weaker-than-expected dereification outcome (IDI_{max} smaller than predicted based on practice hours, though still significantly higher than novice meditators). This might be due to the method potentially reifying body parts and the mantra, respectively, as distinct objects of focus to some extent (in Vipassana, practice instructions of the S.N. Goenka style practiced by this cohort encourage constantly moving attentional focus while mentally scanning body parts; and this particular style of Himalayan Yoga involved mantra practice which involves repetition of a phrase within the phonological loop of working memory).

Secondly, the pooled data from Tibetan dereification practices (Trekchö and Guhyasamaja, totaling 33 participants) demonstrated a substantial deviation, significantly exceeding the expected level of dereification based on lifetime practice hours (one-sample t-test; $t = 2.322, p = 0.030$). Both practices are explicitly targeting dereified modes of existential awareness, with Trekchö focusing on piercing through the illusions of reified thoughts, perceptions, and self to reveal the mind's innate emptiness and clarity^{64–66} and Guhyasamaja culminating in the ‘clear light’ stage, which, similarly, involves freedom from conceptual cognition and from

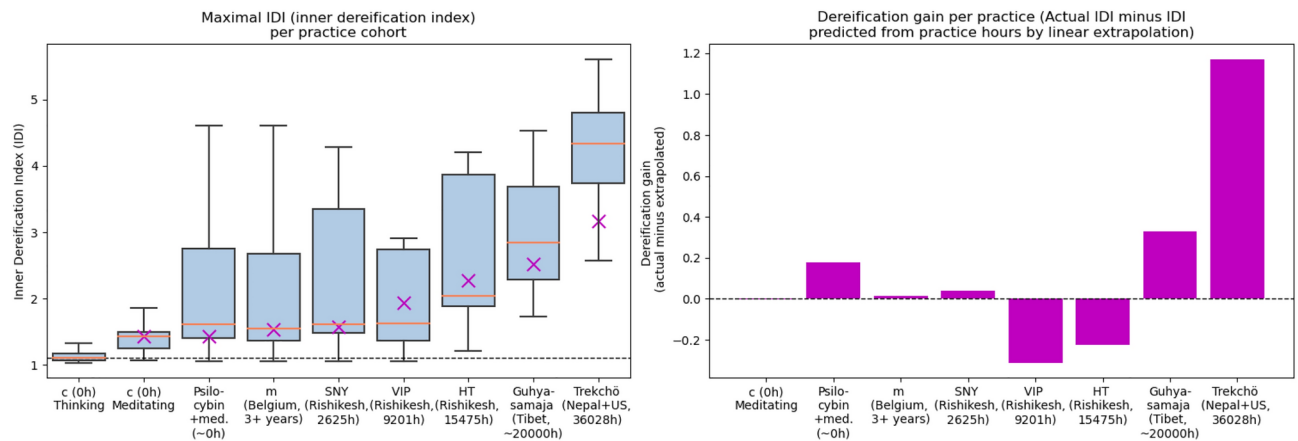


Fig. 2. Inner Dereification Index results across meditator cohorts. Left: IDI_{max} for each cohort in the three datasets (estimating maximal dereification achieved in a session per practitioner—IDI is only a proxy, see “Deviations from default mode network activity as computational markers of meditative practice states” section). Orange lines represent medians, magenta crosses are linear extrapolations based on practice hours, boxes indicate inter-quartile ranges (25–75%), and whiskers extend 1.5x the IQR. Right: Dereification gain, showing the actual median dereification minus the extrapolated median—in other words, how much higher was the actual IDI_{max} compared to what could be expected based on lifetime practice hours (based on linear regression on practice hours and median $IDI_{progress}$). Cohort abbreviations: c—controls (zero hours of meditation experience), Psilocybin+med (individuals with zero or negligible meditation experience doing mindfulness on psilocybin), m—meditators (recruited in Belgium by⁶², 3+ years of experience), SNY—Isha Shoonya Yoga meditators (recruited in Rishikesh by⁶³, 2,625h experience), HT—Himalayan Yoga Tradition (recruited in Rishikesh by⁶³, 15,475h of meditation experience), Guhyasamaja (20,000h experience) and Trekchö, Dzogchen Buddhist meditators (36,028h lifetime practice hours).

grasping for true existence of reified phenomena^{67,103}. These practices exhibited the highest estimated Inner Dereification Index (IDI) across all cohorts, with a median IDI of 4.338 (SD = 1.089) for Trekchö practitioners and 2.846 (SD = 1.150) for Guhyasamaja practitioners, compared to a median IDI of 1.471 (SD = 1.093) in other meditator cohorts. The difference between the Tibetan and non-Tibetan cohorts was highly significant (Mann-Whitney U test; $U = 222.0$, $p < 0.001$), suggesting that these practices, which directly target the dissolution of reified thought and self, lead to markedly higher levels of dereification.

Additionally, the psilocybin cohort, despite involving different mechanisms and phenomenology, also significantly exceeded the expected dereification level for novice meditators (one-sample t-test; $t = 5.549$, $p = 2.067 \times 10^{-7}$). Psilocybin’s ability to deactivate the default mode network and attenuate the sense of self^{38,104,105} may contribute to its pronounced effect on dereification. Finally, emotional states induced by video stimuli yielded the lowest IDI values among all cohorts (median IDI 0.632 - see [Supplementary Information](#)), hinting at emotional states experienced in normal consciousness by untrained individuals without meditative experience not involving any substantial degree of dereification.

These findings collectively provide strong preliminary support for the IDI as a potential proxy for dereification across various contexts, including advanced meditative practices and altered states of consciousness induced by substances like psilocybin. The ability of the IDI to discriminate between different levels of dereification across such diverse cohorts highlights its potential utility as a robust measure in the study of meditative states and related phenomena.

Prediction accuracy: meditation vs. mind wandering

The $IDI(x, \mathcal{X})$ can be calculated in real-time for each short EEG window during a meditation session, allowing for a more granular analysis than a single session-wide metric like IDI_{max} (refer to Eq. 6). We process 3-second EEG windows as outlined in “Methods” section and present the average progression of IDI scores throughout meditation sessions (Fig. 3 left).

The progression curves of individual meditators (faint colored curves) are rarely linear (consistent with phenomenological reports especially by novice practitioners). When averaging over a hundred within-subject score progressions, 0–1 normalized on both axes (due to differing durations across cohorts, and varying IDI_{max} reached), the average tends to increase throughout the session (thick blue line), as can be expected from effective practices. (In contrast, the average IDI score progression tends to very slightly decrease during the mind wandering blocks, at a slope of -0.0011 to 95% confidence interval $[-0.0012, -0.0009]$ —when fitting a linear model). In contrast to the clear averaged trend in meditation, when applying our method to emotional states induced by video clips, no clear progression trend is observed (see [Supplementary Information](#)).

We also assess the model’s ability to differentiate meditation from mind wandering using a tenfold cross-validation after randomization. In each validation fold, the model accesses up to 1 min of mind wandering EEG data, equivalent to 20 data points corresponding to 3-s windows each, as detailed in “Methods” section.

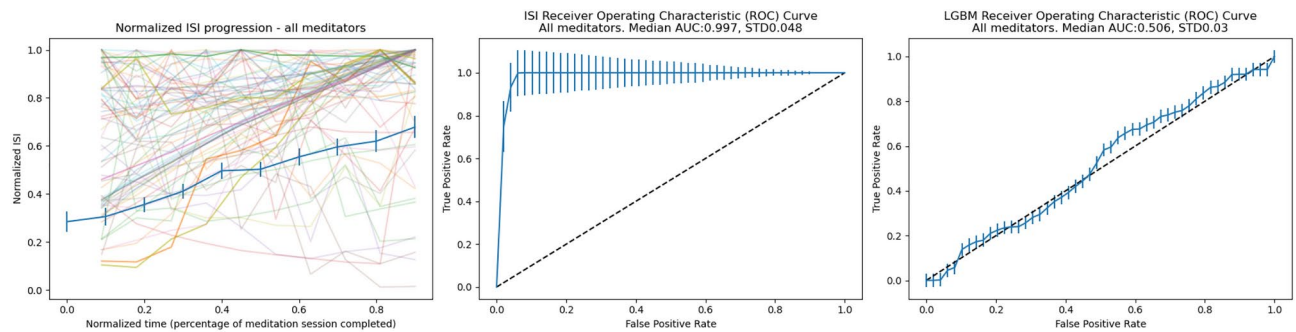


Fig. 3. Inner Dereification Index progression, and classification performance, during meditation sessions. Left panel: averaged 0–1 normalized IDI score progression during a meditation session (models trained on 1-min subsets of EEG recordings of mind-wandering). Middle panel (*one-class classifier*): receiver operating characteristic (ROC) curves of IDI scores specific to individual meditators (trained using 10-fold cross validation: model shown 90% of the data and tested on 10% held-out data), averaged across all meditators. The dotted black line corresponds to random chance performance. Right panel (*binary classifier*): ROC curves of gradient boosted decision tree classifiers (LightGBM) trying to generalize across all meditators, trained using group 10 fold cross validation (LightGBM model trained on data of 90% of meditators, and applied to the held-out 10%).

The IDI model demonstrates excellent performance on held-out test sets, with a median ROC AUC of 0.996. This high accuracy indicates that IDI is a reliable quantitative measure for identifying meditation practice states, effectively distinguishing between mind wandering and meditative states across a majority of EEG time windows (without the need for target state data).

We also report results of a two-class LightGBM⁸⁹ classification model trained on other meditators' data (which requires data available both from meditative and mind wandering states, from a training cohort of individuals), which attempts to generalize learned patterns across individuals. Results are shown in the right panel of Fig. 3. The performance of such a model, trained using 10-fold cross validation, lies at 0.506 median ROC AUC (with 0.030 standard deviation), suggesting that it is not possible to robustly generalize cortical signatures of meditation states across participants. (Note: The performance of binary classifiers is likely to be higher when using substantial training data from both states of the same participant, rather than a training cohort of other participants. An example set-up could record several sessions of meditation, as well as mind wandering, from the same individual, and subsequently train the model on a subset of sessions of both kinds of states, and evaluate its performance on a held-out subset. This setup would place a much higher burden of time requirement on each participant before a model can be learned, and would require participants to be able to enter desirable meditation states already, precluding the use of this procedure with novice meditators or in settings where recordings of a target states are unavailable, as well as traditions eschewing a particular target or goal state).

Discussion

The increasing evidence of meditation's benefits has spurred interest in effectively quantifying meditative states. Quantitative measures of desirable cortical states, such as meditation or positive affect, may be useful across a wide variety of applications, including descriptive studies, neurofeedback (real-time feedback from brain activity in order to reinforce desirable brain function, or teach self-control of brain activity), retrospective or real-time evaluations of intervention effectiveness, comparisons of cohorts or methods, progress tracking, and potential therapeutic applications of feedback-assisted generation of positive states or contents of consciousness.

Previous work has identified multiple electrophysiological correlates of meditative states, but lacked the predictive accuracy to cleanly distinguish meditative from non-meditative states or quantify a meditation-relevant aspect (see Introduction). Many of these works attempted to generalize the proposed biomarkers across all participants, performing inter-subject analyses, neglecting or ignoring individual differences in cortical oscillatory activity and neural architecture.

In the sections above, we have described a novel method (termed IDI) built on the idea of participant-specific modeling, trainable on a small amount of EEG data from a mind wandering state (rather than requiring data on the target state), and have presented evidence that it may be a useful proxy for the dereification of meditative practice states. Based on these results, IDI seems to have the following desirable properties:

- ability to approximate dereification in meditation with sufficient accuracy to stratify participant cohorts based on their meditation experience (Fig. 2),
- ability to distinguish meditation practices directly and effectively facilitating the highest ranges of dereification (Fig. 2),
- ability to indicate progress during a single meditation session, and accurately distinguish moments of meditation from moments of mind wandering (Fig. 3), and
- applicability to settings where there is no data available from a desirable state. In addition, the data presented above strongly suggests that personalized, individual models of cortical activity can significantly outperform

models that attempt to learn generalizable patterns across a large number of subjects (possibly due to large inter-individual differences between cortical oscillatory patterns associated with meditative states).

Implications and future work

The possibility to quantify meditative states with very light prerequisites (1 min EEG recording of mind wandering) opens the door towards applications involving assistive technology such as neurofeedback. Future work could investigate whether such feedback might be of benefit to meditation practitioners (preliminary evidence for the potential effectiveness of neurofeedback to facilitate meditation has been presented by^{106–109}; such work may be enhanced by more accurate measures of meditative states such as the one presented here).

A straightforward extension of the present work might include the use of multiple undesirable states to refine the measure of a target states (and the potential neurofeedback guidance leveraging such a measure). For example, there are several contemplative traditions which emphasize additional criteria of desirable states of consciousness in addition to lack of mind wandering or negative affect, such as compassion or love, spaciousness, boundlessness or other spatial properties of the experiencer, non-conceptual or non-symbolic or non-particularizing perception, or the cessation of particular contents of experience such as time perception (or even cessation of all experiencing)^{78,110–115}.

Having shown the ability of IDI to constitute a proxy for the dereification aspect of meditative practice states by means of evaluating ‘outlierness’ compared to mind wandering states, there is reason to suspect that the same methodology may work for some of these other aspects of meditation, thus opening the door towards more easily quantifiable research results, as well as potential technological aids (e.g. neurofeedback) to learning meditative skills.

Finally, future work may investigate applicability of IDI—and related one-class models—to therapeutic settings, where patients may find it challenging to generate clean desirable brain states due to present psychopathology (for instance, potentially aiding the generation of positive affect for depressive patients, or certain relational feelings for patients with attachment disorders, using biofeedback). Using similar methods, patients may be able to benefit from neurofeedback to learn a measure of control in navigating towards beneficial states ([Supplementary Information](#)).

Conclusion

The Inner Dereification Index (IDI) introduced in this paper offers a new and highly accurate approach to quantifying aspects of meditative states, and argued that it can serve as a proxy for dereification in meditation. We have also shown its applicability across a wide range of participant cohorts for measuring or classifying aspects of meditative practice states based on EEG data. The strong predictive performance (0.996 AUC) and stratification ability ($F(8, 211) = 7.38, p < .001$) of these models lends credence to the possibility that subject-specific, personalized models (rather than models attempting to generalize across all humans), and one-class approaches (rather than models requiring data from both meditative and mind wandering states) may have significant advantages over traditional biomarkers and statistical methods for the purpose of quantifying and classifying states of consciousness.

The demonstrated utility of IDI across diverse contexts, from advanced Tibetan meditation practices to psilocybin-induced mindfulness, reinforces its potential as a bridge between traditional contemplative techniques and modern therapeutic approaches involving altered states of consciousness. Future work will focus on validating IDI across a wider range of contexts and refining its application in therapeutic and neurofeedback scenarios.

Data availability

The datasets analysed during the current study are available in the following OSF, Zenodo and GitHub repositories: https://osf.io/3uszv/?view_only=b260d7611a664e87a3c291bcd9367a7b, <http://doi.org/10.5281/zenodo.57911>, <https://github.com/JacobHYoung/Contrasting-EEG-Derived-Entropy-and-Neural-Oscillations-During-Six-Meditation-Practices.git>, <https://osf.io/uy83t/> and <https://github.com/IoBT-VISTEC/EEG-Emotion-Recognition-INTERFACES-datasets>. The Guhyasamaja dataset is available from the authors⁶⁷. No additional data was collected or generated as part of the present work.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent

We have used online open-access data made available by^{62,63}, and⁵ for the experiments, who have ensured appropriate approvals, consent and data availability.

Additional information

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Correspondence and requests for materials should be addressed to T.M.

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